

# YOONJI CHOI

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## EDUCATION

### New York University Courant Institute

Master of Science in Computer Science

Relevant Coursework: Machine Learning, Realtime & Big Data Analysis, Vision Meets ML

New York, New York

Expected May 2027

GPA: 3.45

### Stony Brook University

Bachelor of Science in Applied Mathematics and Statistics

Minor in Computer Science | Honors: Cum Laude, Dean's List

Stony Brook, New York

Feb 2022 – May 2025

GPA: 3.63

## SKILLS

**Programming Languages:** Python, SQL, Java, R, C, MATLAB, SAS

**ML & Deep Learning:** PyTorch, SHAP, Statistical Modeling

**Data & Distributed Systems:** Hive, Trino, Dataproc, Hadoop, HDFS

**Infrastructure:** HPC (SLURM, Singularity), OpenMP, MPI, Streamlit

## PROJECTS

### Vision Model Training (CNN, ResNet, DETR)

Mar 2026 – May 2026

- Trained and evaluated CNN, ResNet, and Detection Transformer (DETR) models on large-scale image datasets using PyTorch and GPU acceleration.
- Developed end-to-end training pipelines on the NYU Greene HPC cluster using data augmentation, mixed precision, and learning rate scheduling, and analyzed training dynamics under architectural modifications.
- Implemented object matching methods related to DETR and analyzed model behavior across architectures through debugging and performance evaluation.

### pico-LLM: Transformer FFN Interpretability Analysis

Nov 2025 – Dec 2025

- Modeled and compared k-gram MLP, LSTM, and Transformer architectures in PyTorch within a collaborative GitHub workflow.
- Designed hypothesis-driven experiments using an activation logging pipeline, finding FFN activations progress from lexical to discourse-level representations.

### Traffic Accidents Causal Analysis

Nov 2025 – Dec 2025

- Built a large-scale ETL pipeline integrating collision, traffic volume, ride-hailing, and weather datasets using Dataproc, Hive, and Trino.
- Applied Poisson, Negative Binomial, and Random Forest modeling ( $R^2 \approx 0.41$ ) to quantify crash risk drivers, finding snow conditions elevate hourly crash risk by ~25%.

### Olist Return Analysis

Feb 2025 - May 2025

- Created an end-to-end pipeline (preprocessing, EDA, statistical modeling) on 100K+ e-commerce transactions under extreme class imbalance (~0.48% return rate), incorporating NLP analysis of customer reviews.
- Developed predictive models with SHAP-based interpretability and deployed a Streamlit dashboard visualizing review sentiment, purchase behavior, and return-rate relationships.

## RELATED EXPERIENCE

### Graduate Research Assistant – New York University

May 2026 – Present

Advisor: Prof. Grace Lindsay

- Conducting research on context-dependent recognition in vision-language models, reviewing literature on shape/texture bias and multimodal conflict resolution.
- Planning prompt-based task experiments and perceptual analysis to evaluate how contextual cues influence VLM recognition behavior.

### Undergraduate Research Assistant – Stony Brook University

Jan 2025 – May 2025

Advisor: Prof. Hyunkyung Lim

- Independently developed and optimized parallel PDE/ODE solvers using OpenMP and MPI, improving runtime by up to 1.6× across configurations and delivering performance improvements with minimal day-to-day supervision.
- Validated numerical accuracy against analytical solutions across serial and parallel configurations, maintaining reliability standards in a self-directed research environment.